

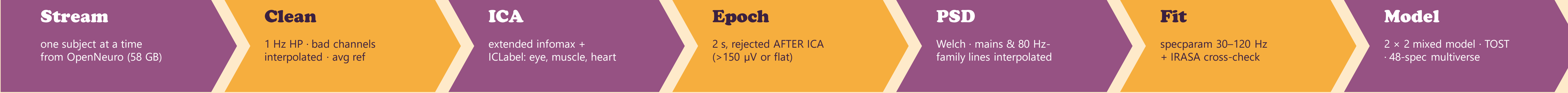
The meditation “gamma effect” has no gamma in it

A specparam re-analysis of 94 people in OpenNeuro ds003969. The famous “meditation gamma” was never an oscillation: it is aperiodic background. The one effect that survives correction: novices’ spectra steepen during ten quiet minutes of breath-counting; meditators’ hold steady.

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94 people, 3 traditions + controls
341 eyes-closed 10-min recordings
6 % of recordings with any 60–110 Hz peak

Trait meditators vs controls: not significant after correction.
State thinking → meditation: exponent steepens, Holm p = .02.
State × trait the steepening is the controls’: Holm p = .048; IRASA .018.



Pipeline: open, unit-tested, one command (python scripts/run_all.py). Analysis plan committed to git before the confirmatory run.

1 · The problem

Two things, one number

Braboszcz, Hahusseau & Delorme (2017) reported more 60–110 Hz “gamma” power in long-term meditators than controls, scaling with years of practice, one of the most-cited EEG findings on meditation.

But band power adds two things: **periodic** activity (a real oscillation, a peak in the spectrum) and the **aperiodic 1/f^α** background (offset + exponent). A flatter or higher background raises “gamma” with no rhythm at all, and the exponent is exactly what drowsiness, anaesthesia and scalp muscle move.

Two newer studies disagree about that background. Pascarella ... Jerbi 2025 (MEG, state): it flattens during meditation. Biswas ... Ray 2026 (EEG, trait): it is steeper in meditators. Nobody had tested both in one dataset.

2 · The data

ds003969: 98 recorded, 94 analysed, fully crossed 2 × 2, CC0

- Rishikesh, Biosemi 64 ch, 1024 Hz, eyes closed.
- Four 10-min blocks each: meditation / thinking / meditation / thinking, order counterbalanced. Controls count breaths in the “meditation” blocks.
- The same people give a TRAIT contrast (meditators vs controls) and a STATE contrast (meditation vs thinking). Four undocumented “tm” participants excluded.

Group	n
Controls	31
Himalayan Yoga	24
Isha Shoonya	20
Vipassana	19

341 usable recordings of 376; 14 dropped before fitting (>16 dead channels or <60 s clean data), 21 with fit R² < 0.9. Same exclusion rate in both groups.

3 · The method

Separate the peak from the background, then model both

log P(f) = offset − χ · log f + Σ Gaussian peaks specparam 2.0 · 30–120 Hz · parieto-occipital

Five measures per recording: **naive gamma power** (the 2017 measure), **periodic gamma power** (peaks only), **aperiodic gamma power** (background only), **exponent χ** and **offset**.

Model: measure ~ meditator × condition + block order + (1 | subject). Holm-corrected within contrast; nulls get TOST equivalence tests (±0.5 SD).

Cross-check: IRASA re-estimates the exponent by a different algorithm (r = 0.96 with specparam).

4 · Validation

Known ground truth: what does each measure report?

	what each measure reports (Cohen’s d, meditation – thinking)		
	naive 60–110 Hz band power	periodic gamma power	aperiodic exponent
real gamma peak	+0.6	+38	0
flatter exponent	+3.0	0	-3.3
higher offset	+1.1	-0.2	0
peak + exponent	+3.3	+44	-3.1
no change	0	0	0

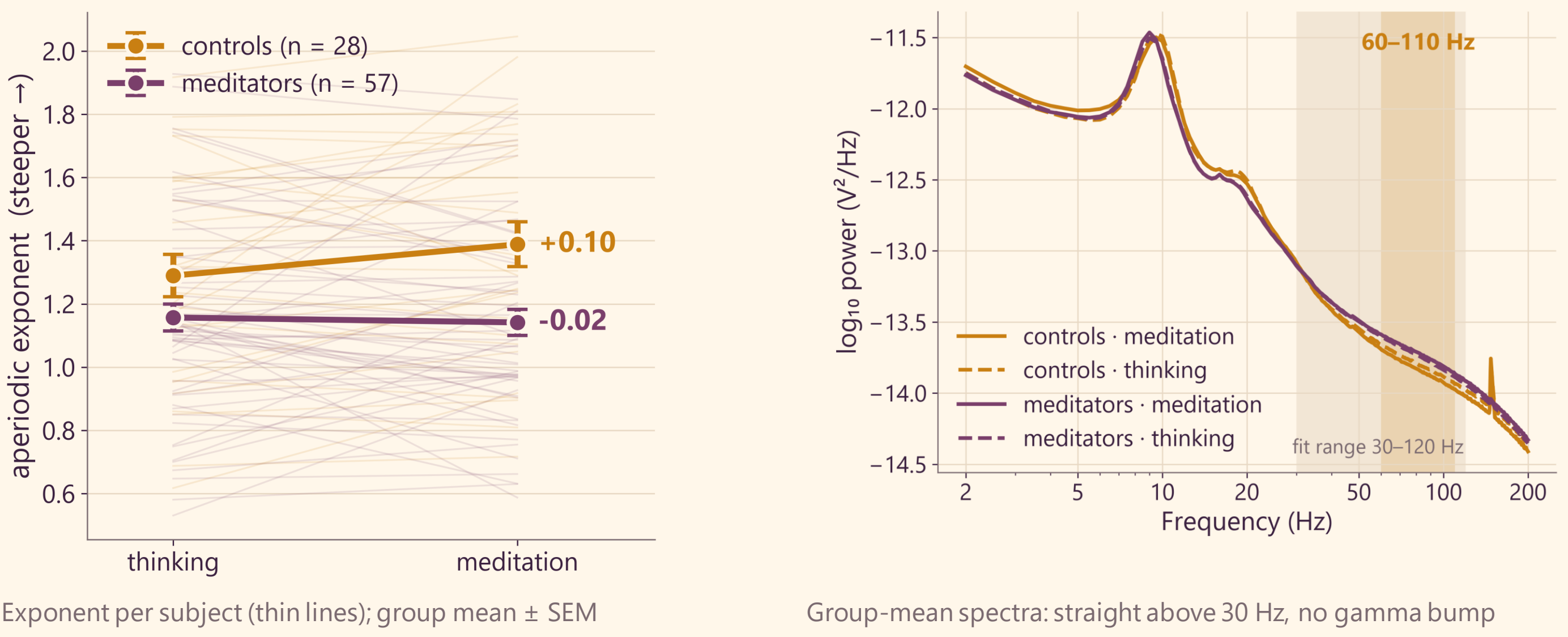
d ≈ 3 reported by naive band power when only the exponent changed

d ≈ 0.6 reported by naive band power for a genuine gamma peak

Simulated paired thinking/meditation spectra under five scenarios, run through the identical fitting code. Naive band power reports an “increase” in every non-null scenario and is five times more sensitive to the background than to the oscillation it claims to measure. The decomposed measures attribute each scenario correctly. These criteria are unit tests.

5 · Main result

The 2 × 2 on the aperiodic exponent



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recordings with a countable 60–110 Hz peak (mostly at fit edge)

+0.10

controls: exponent steepens thinking → meditation, p = .04

−0.02

meditators: no change (equivalent to zero, TOST p < .001)

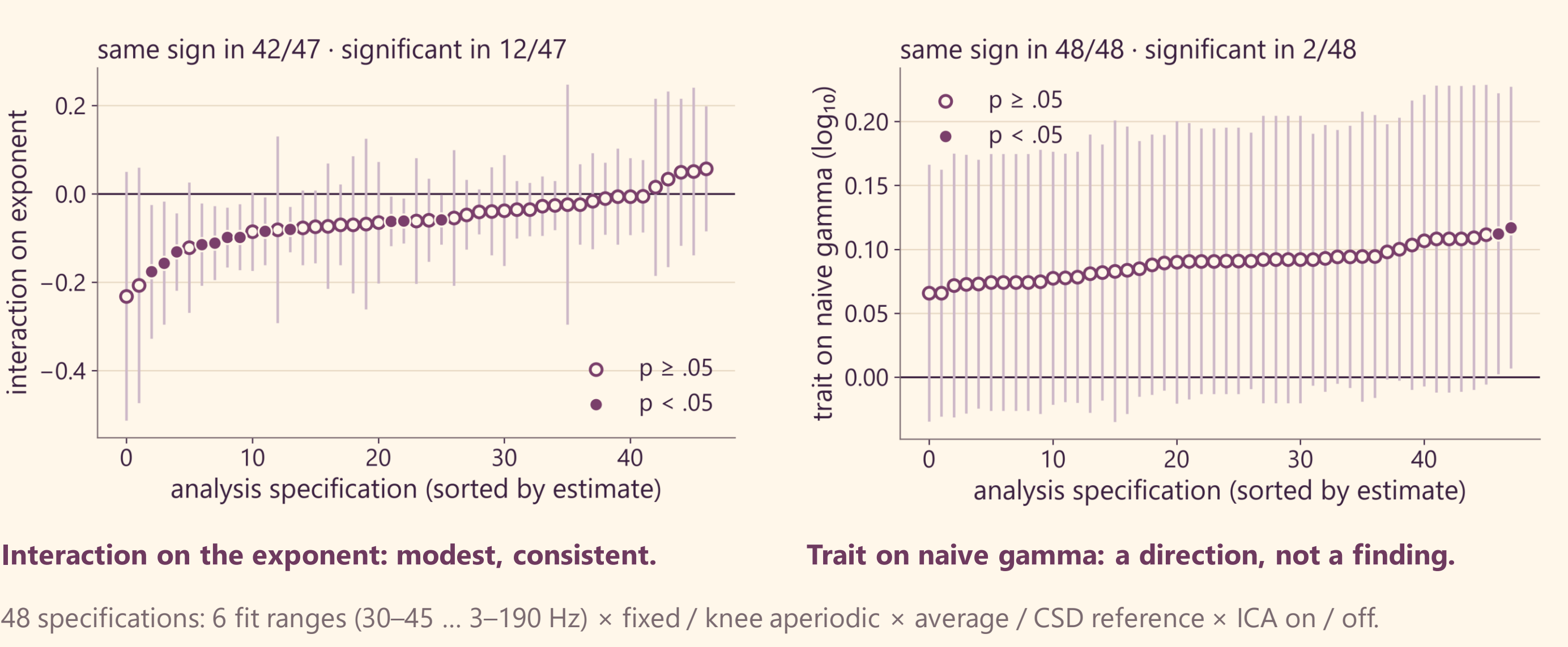
No periodic gamma. Periodic 60–110 Hz power is ≈ 0 for everyone; the few fitted peaks sit where the fitter absorbs edge curvature. Whatever “gamma” is here, it is the aperiodic background.

Trait (H1/H4): not a replication. Meditators vs controls: naive gamma +0.09, exponent −0.12 (flatter, not steeper as Ray reports). Same direction as 2017, but p ≥ .30 after Holm correction and significant in 2 of 48 specifications.

State × trait: the result that survives. The exponent steepens from thinking to meditation (Holm p = .02), but only in controls. Interaction Holm p = .048, IRASA p = .018, offset agrees (p = .01), same sign in 42 of 47 specifications; unchanged after adjusting for two scalp-muscle proxies (−0.109 → −0.103).

Reading. A steeper exponent is the best-replicated EEG signature of lower arousal. Ask a novice to sit eyes-closed counting breaths for ten minutes and their spectrum drifts toward a drowsier state; a trained meditator’s holds steady.

6 · Sensitivity



Interaction on the exponent: modest, consistent. 48 specifications: 6 fit ranges (30–45 ... 3–190 Hz) × fixed / knee aperiodic × average / CSD reference × ICA on / off.

Trait on naive gamma: a direction, not a finding.

7 · Controls

Muscle

Two EMG proxies per recording (temporal 70–200 Hz power before ICA; ICLabel muscle variance) added as covariates: the interaction is unchanged (−5 %) and the trait effect grows ~20 %. And controls show LESS high-frequency power during meditation, the opposite of muscle tension.

Method

IRASA, an unrelated algorithm, reproduces every specparam effect (interaction p = .018 vs .013). The two exponents agree at r = 0.96.

Order & QC

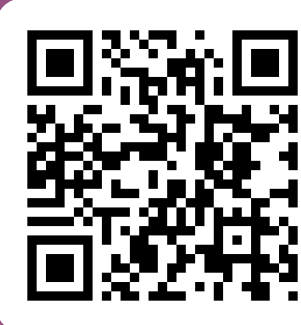
Block order is in the model; half the sample started with each condition. Automated QC caught an 80 Hz equipment line in ~50 % of recordings, dead frontal electrodes in one subject, and a truncated download.

8 · What this cannot show

Sensor EEG cannot localise the change; EMG adjustment bounds but does not eliminate muscle. Arousal is inferred from the exponent, not measured. Cross-sectional: practice may cause the trait difference or different brains may take up practice. The plan was fixed during a 20-subject pilot and interim results were viewed twice; hypotheses were not changed. Eyes closed only, so Ray’s eyes-open result is not directly reproduced.



Data OpenNeuro ds003969 (CC0)



Code, analysis plan, tables github.com/cation21/Gamma

Braboszcz, Hahusseau & Delorme 2017 PLOS ONE · Donoghue et al. 2020 Nat Neurosci · Pascarella ... Jerbi 2025 Neurosci Conscious · Biswas ... Ray 2026 Imaging Neurosci · Muthukumaraswamy 2013 Front Hum Neurosci · Wen & Liu 2016 Brain Topogr · Gerster et al. 2022 Neuroinformatics · Lendner et al. 2020 eLife